Table 1: The training schedules used in our experiments.

Tokens	**Global Batch Size (tokens)**	**Iterations**	**Learning Rate**	**Warmup**	**Cooldown (20%)**
100B	4.12M	23,842	4×10^{-3}	8,334	4,768
300B	4.12M	72,661	4×10^{-3}	25,000	14,532

Table 2: **open-sci-ref** [nezhurina2025openscіref] model architecture and scales. We used tied embedding weights in all experiments.

Parameters (B) (Non-Emb + Emb)	**Layers**	**Hidden**	**Heads**	**FFN Hidden**	**Memory**	**FLOPs**
$0.35 + 0.05 = 0.40$	22	1024	16	3840	2.88 GB	2.4×10^{9}
$1.61 + 0.10 = 1.71$	24	2048	32	8192	9.884 GB	1.0×10^{10}

Table 3: Hyperparameters used for OT3 (OpenThoughts-3, 30k samples) SFT runs. All runs share the same recipe, and only the attention backend and chat template change per architecture.

Hyperparameter	Value
Dataset	`open-thoughts/OpenThoughts-3` (30k samples)
Stage / fine-tuning type	SFT, full-parameter
Trainer / framework	LLaMA-Factory
Distributed strategy	DeepSpeed ZeRO-3
Precision	BF16
Liger Kernel	enabled
Attention backend	FA2 (Llama/Qwen) / SDPA (opensci-arch)
Sequence length (cutoff_len)	16,384
Per-device batch size	1
Gradient accumulation steps	4
World size	32 H100 (8 nodes $\times$ 4)
Effective global batch size	128
Optimizer	AdamW (LLaMA-Factory default)
Learning rate	4×10^{-5}
LR schedule	Cosine
Warmup ratio	0.1
Epochs	5
Group by length	true
Seed	42

Table 4: Single-run GPU-hours for OT3 (30k) SFT. All runs: 32 H100 (8 nodes × 4 GPUs). Difference between groups is the attention backend: FA2 (Llama/Qwen) is ∼2× faster than SDPA (opensci-arch).

Base model	Attention	GPU-hours	Training
MixtureVitae-1.7B	SDPA	106	OT3 30K SFT
MixtureVitae-1.7B + SFT-Tulu3	SDPA	106	OT3 30K SFT
MixtureVitae-1.7B + DPO-Tulu3	SDPA	105	OT3 30K SFT
Comma0.1-1.7B + SFT-Tulu3	SDPA	105	OT3 30K SFT
Comma0.1-1.7B + DPO-Tulu3	SDPA	158	OT3 30K SFT
FineWeb-1.7B + SFT-Tulu3	SDPA	106	OT3 30K SFT
Nemotron-1.7B + SFT-Tulu3	SDPA	106	OT3 30K SFT
Nemotron-1.7B + DPO-Tulu3	SDPA	106	OT3 30K SFT
Qwen3-1.7B-Base	FA2	60	OT3 30K SFT
Qwen3-1.7B-Base + SFT-Tulu3	FA2	57	OT3 30K SFT
Qwen3-1.7B-Base + DPO-Tulu3	FA2	57	OT3 30K SFT
Qwen2.5-1.5B	FA2	53	OT3 30K SFT
Qwen2.5-1.5B + SFT-Tulu3	FA2	53	OT3 30K SFT
Qwen2.5-1.5B + DPO-Tulu3	FA2	52	OT3 30K SFT
SmolLM2-1.7B	FA2	48	OT3 30K SFT
SmolLM2-1.7B-Instruct	FA2	48	OT3 30K SFT
SmolLM2-1.7B + SFT-Tulu3	FA2	44	OT3 30K SFT
SmolLM2-1.7B + DPO-Tulu3	FA2	47	OT3 30K SFT
Total (18 runs)		**1417**	

Table 5: Single-run GPU-hours for long-context extension SFT. All runs: 8 GH200 (2 nodes × 4 GPUs), 1 epoch (≈16k tokens/seq, YaRN RoPE scaling), SDPA attention.

Base model	Attention	GPU-hours	Training
MixtureVitae-1.7B, Comma0.1-1.7B, DCLM-1.7B, FineWeb-Edu-1.7B and Nemotron-HQ-1.7B, each	SDPA	12	Long-ctx SFT (16k)

Table 6: Pretraining token budget for each model and its ratio relative to MV (300B tokens). External open-weights references use substantially more pretraining compute than MV.

	{MV,Comma0.1,FW-Edu,Nemotron}	**SmolLM2**	**Qwen2.5-1.5B**	**Qwen3-1.7B**
# Tokens (T)	0.3	11	18	36
Budget Ratio	1.0	36.7	60.0	120.0

Table 7: **Protocol A, the unified evaluation protocol.** Used for Phase 1, the 300B front-loading factorial and the mixture ablations. All results under this protocol are reported as mean and sample standard deviation over decoding seeds 42, 1234, and 2025. GSM8K is scored as under Protocol B (Table 8), by exact match on the last number of the answer, the LM Evaluation Harness's flexible-extract filter. The suite average is the unweighted mean of the eleven tasks, a diagnostic to read alongside the per-task scores. It weights each domain by its number of tasks: mathematics at 45%, code at 27%, science at 18% and instruction following at 9%.

Abbr.	**Task**	**Domain**	**Ref.**	**Metric**	**#Shots**	**Decoding**	**Max new toks**
A24	AIME24	Mathematics	[aime2024]	Accuracy	0	Greedy	32,768
A25	AIME25	Mathematics	[aime2025]	Accuracy	0	Greedy	32,768
AMC	AMC23	Mathematics	[maa2023amc]	Accuracy	0	Greedy	32,768
M500	MATH500	Mathematics	[hendrycks2021math]	Accuracy	0	Greedy	32,768
GSM8K	GSM8K	Mathematics	[cobbe2021training]	Exact match (flexible extract)	5	Greedy	2,048
HE	HumanEval	Code	[chen2021evaluating]	`python_pass@1`	0	Greedy	2,048
LCB	LiveCodeBench (v5)	Code	[jain2024livecodebench]	Pass@1	0	Greedy	32,768
MBPP	MBPP	Code	[austin2021program]	Functional Pass@1	3	Greedy	2,048
GPQA	GPQA-Diamond	Science	[rein2024gpqa]	Accuracy	0	Greedy	32,768
JEE	JEEBench	Science	[arora2023have]	Accuracy (JEE score)	0	Greedy	32,768
IFE	IFEval	Instruction following	[zhou2023instruction]	Instruction-level accuracy (strict)	0	Greedy	2,048

Table 8: **Protocol B, the reasoning post-training protocol.** Used for every OpenThoughts3-30K experiment, namely the endpoints of Table **??**, the pretraining ablation of Table **??**, and the grid of reasoning corpora of Tables **??** and **??**. Every benchmark is sampled at temperature 0.7, with one answer per question in each of three runs, and results are the mean and SD over the runs.

Abbr.	Task	Ref.	Metric	#Shots	Decoding	Max new toks
A24	AIME24	[aime2024]	Accuracy	0	Sampling (T=0.7)	32,000
A25	AIME25	[aime2025]	Accuracy	0	Sampling (T=0.7)	32,000
AMC	AMC23	[maa2023amc]	Accuracy	0	Sampling (T=0.7)	32,000
M500	MATH500	[hendrycks2021math]	Accuracy	0	Sampling (T=0.7)	32,000
GSM8K	GSM8K	[cobbe2021training]	Exact match (flexible extract)	5	Sampling (T=0.7)	16,000
HE	HumanEval	[chen2021evaluating]	Pass@1	0	Sampling (T=0.7)	16,000
LCB	LiveCodeBench (v5)	[jain2024livecodebench]	Pass@1	0	Sampling (T=0.7)	16,000
MBPP	MBPP	[austin2021program]	Pass@1	0	Sampling (T=0.7)	16,000
GPQA	GPQA-Diamond	[rein2024gpqa]	Accuracy	0	Sampling (T=0.7)	32,000
JEE	JEEBench	[arora2023have]	Accuracy (JEE score)	0	Sampling (T=0.7)	32,000
IFE	IFEval	[zhou2023instruction]	Instruction-level accuracy (strict)	0	Sampling (T=0.7)	16,000

Table 9: Standard language understanding evaluation benchmarks used in our result tables. All tasks follow the default the LM Evaluation Harness [eval-harness] configurations, with prompt-example seed 1234 and no chat template. Normalized accuracy is the harness's `acc_norm`, which scores each answer by its log-likelihood per byte. MMLU is the harness's overall score, the mean over all 14,042 questions of its 57 subjects.

Abbr.	Task	Ref.	Metric	#Shots
MMLU	MMLU	[hendrycks2020measuring]	Accuracy	5
HellaS	HellaSwag	[zellers2019hellaswag]	Normalized accuracy	10
CSQA	CommonSenseQA	[talmor2019commonsenseqa]	Accuracy	10
ARC-C	ARC-Challenge	[clark2018think]	Normalized accuracy	10
ARC-E	ARC-Easy	[clark2018think]	Normalized accuracy	10
PIQA	PIQA	[bisk2020piqa]	Normalized accuracy	10
BoolQ	BoolQ	[clark2019boolq]	Accuracy	10
WinoG	Winogrande	[sakaguchi2020winogrande]	Accuracy	0
OBQA	OpenBookQA	[mihaylov2018openbookqa]	Normalized accuracy	0
COPA	COPA	[copa]	Accuracy	0
LAMBADA	LAMBADA	[paperno2016lambada]	Accuracy	0